# Dual-ligand fluorination



## 01 / Experiment

Use the bench sheet. Staff weigh/coarse-load chemicals, handle caps/labware and start Python locally; the robot measures liquids. Keep generated CSVs with their protocol.

### Chemistry and design

**Pyridine-2-sulfonyl fluoride → 2-fluoropyridine.** Nominal per vial: 25 µmol SM, 1.5 µmol each ligand, 3 µmol Pd(COD)(DQ), 2.5 µmol naphthalene; 250 µL toluene, 95 °C, 500 rpm, 4 h under air. No base/external fluoride. Solvent addition defines nominal concentrations; solute displacement/contraction are accepted approximations.

R1: **48 pairs covering all 30 ligands**, official 24-consensus +24-Sobol design; L07 excluded. Regenerate stocks/protocols/maps. R2: **22 pairs +L17 tBuBrettPhos at 12 mol% total ligand**; L03 at most four wells. No repeats. Higher yield identifies a preliminary candidate; batch effects, reproducibility and cooperativity need later matched controls.

### Robot stock preparation

1. Enter **actual masses/assays** in `weighed_charges_template.csv`. Weigh ligands in the glovebox; keep bottles there. Pd: **4 A3/B3/C3, 22.8 mg each; A4, 21.6 mg** (≤90 mg; keep 10 mg). Naphthalene: **60 mg in 1 C2**; `IS_MASTER` empty. Under air, the robot adds solvents, mixes and prepares reaction/analytical stocks. Preparation is inside campaign time.
2. **Live release is blocked:** bottom aspiration in the configured 15 mL tubes requires unverified nozzle entry; the conservative reach check fails. Obtain actual short, solvent-compatible stock containers and matching rack JSON, then regenerate and check reach/capacity. Follow carrier/raw-load CSVs; no invented glass-vial dimensions.
3. Run `prepare_stocks.py`: **p300 single left, p1000 single right**; tips 5 /8-9 respectively. Carriers occupy 3/6 temporarily; working racks 1/4. The supplied horizontal 8 ×30 mL reservoir in 2 is **single-channel only**. Robot prepares analytical matrix/masters/working stocks too.
4. Cap/store reserve carriers and the prepared analytical rack. Leave working tubes in mapped positions. Change right pipette to **p300 eight-channel GEN2**, replace tips 8/9 with 300 µL racks using the local App installation/calibration workflow. Keep left p300 single.

| Slot | Reaction-stage use |
|---|---|
| 1 / 4 | 2 mL working tubes; max 1.95 mL. Ligands at CSV addresses; 500 mM IS **1 C2**, SM/IS **4 D2**, Pd **4 A3/B3/C3/A4**; prep acid **4 D3**. Retain RockPhos, SM/IS and all four Pd tubes. |
| 2 | New EMPTY 12-channel reservoir. Robot prepares workup fluids here during the hold; no staff premixing/refills. |
| 3 / 10 | Reaction block at pipetting / heater-shaker positions; manual moves. During holds, 3 temporarily takes raw solvents. |
| 6 | Calibration plate during R1; later a fresh final sample plate per round. Three compatible 96-well LC plates total. |
| 5 / 8 / 9 | Single / multi / multi tips. Keep 7 and 11 empty. |

Confirm block/adapter/latch, offsets, <=45 mm cap clearance and liquid immersion with **1.5 mm bottom clearance**. Confirm LC plate/carrier/seal/needle depth and the requested full 1.7-min method/cycle, UV channel, RTs/LOQs. Remaining physical setup facts.

### Run R1

Start R1 Python with **48 NEW EMPTY vials** in the block at 3. Robot doses **25 µL SM/IS +75 µL each ligand +75 µL Pd**. Retain SM/IS tube **4 D2**, capped between rounds: its initial **1.89 mL** comes from the same nominal 2 mL stock; no additional SM stock or refill. Expected remainder after R1: **689 µL**; R2 requires **at least 680 µL**. Record any loss.

Hand-cap all R1 vials; move/latch at App prompts. Run **95 °C /500 rpm /4 h**. During this hold the robot prepares **14 standards** and quench/diluent. Swap analytical rack into 4 and **uncap its source tubes**; raw reservoir into 3. Seal/move calibration plate 6→HPLC. **Reaction stays on 10.** Restore rack 4 and remove raw reservoir when prompted.

**Start a 4 h external timer at each App hold-start message.** Preparation/pauses count. Code cannot stop heat while App-paused. At deadline, stop heat/shaking through the App; record deviations and do not resume dosing. Logging tolerance does not establish chemical equivalence.

## Workup and final LC plates

1. Move sealed block to 3; **cool vials before opening**. App separately checks module <=30 °C. **Uncover reservoir 2; keep vials capped while robot pre-fills the fresh LC plate in 6 with 980 µL diluent/well.** Then uncap; robot adds **500 µL 9:1 ACN/water +0.15% FA**. Nominal extract: 750 µL, 0.1% acid, 1.25 mL headspace. Neither fluid contains IS.
2. **Leave reaction vials OPEN**; shake **750 rpm /2 min**, heater off, at the locally confirmed splash-free setting. Return block to 3, let bubbles settle briefly and sample while open. No recap/reopen cycle before sampling.
3. Sample promptly after shaking; LC diluent is already in place. Robot uses a fresh tip, two slow same-well prewets, **21 µL aspiration at 1.5 mm**, 1 µL retained reverse surplus and 2 µL air gap. Dispense nominal **20 µL into prefilled LC wells**, noncontact; discard tip. No manual dilution/vial transfer.
4. Recap reaction vials. Move final LC plate **6→cooled shaker 10**, A1 orientation unchanged. Robot shakes **UNCOVERED, heater OFF, 750 rpm /2 min** at the locally confirmed splash-free speed. Reaction block remains in 3.
5. Move LC plate **10→HPLC**, apply compatible pierceable seal and inject directly. R1: **R1-LC01 A1-H6**, 48 samples. R2: **R2-LC01 A1-H2 plus A3-G3**, 23 samples. **H3 is diluent only; never inject.** Follow generated sequence/maps; reaction and LC coordinates differ.

Final sample: 1 mL; nominal maximum analyte **666.7 µM**, naphthalene **66.7 µM**, FA **0.1%**, toluene **0.667%**. Use the calibrated injection volume. Seal after final shaking to limit evaporation. There are four final-sample plate moves plus the calibration-plate handoff.

## R2 and finish

After ALL 48 R1 results pass QC, select 22 pairs +L17 reference. Run `refill_stocks.py` **before R2 dosing**. Remove block/LC plate from 3/6; present ORIGINAL bulk carriers by ID. **Uncap required bulk/working ligand tubes for top-ups.** No new weighing or hand aliquoting. Retain RockPhos; **keep SM/IS and all four Pd tubes capped during refill**. Pd routing: R1 **18/18/12 doses** from A3/B3/C3; R2 **six C3 +17 A4**. Actual mass/assay determines solvent additions and capacity.

Load **23 NEW EMPTY vials** in the SAME block; replace reservoir 2 with a NEW EMPTY one. Uncap required stocks. R2 Python doses **25 µL SM/IS each, then catalysts**; reference receives **150 µL L17**. Hand-cap and start the reaction. During R2 hold the robot makes fresh workup fluids; no new calibration. Repeat workup and final LC analysis.

Cap stock reserves between stages. Robot mixes sources before first use/every eight transfers, without air gaps/blowouts. Same-stock dosing tips may be reused only for validated noncontact transfers. Analytical samples always use fresh tips. Mixing can disperse solids; it does not prove a settling suspension is uniform.

Pause for visible layers, blocked tips, persistent bubbles, air uptake, dripping or tip contact. Do not sample an arbitrary layer, guess a top-up or filter/decant without recording the effect. Reprepare an uncertain analytical dilution robotically and record the replacement; retain rejected measurements. No separate homogeneity study is planned.

Finish after all 23 final injections and quantitative analysis. Save robot JSON logs, raw LC/MS/method/sequence files, actual weighed charges, times and deviations under `local_data/`. Run `python scripts/hte-return-package.py`; send its ZIP privately. It retains code, preparation/source ledgers, inputs, results, Claude records and file hashes; credentials are excluded.

# 02 / Stocks and calibration

**Staff weigh; the OT-2 measures liquids.** Enter actual masses/assays; follow `bulk_carriers.csv`. Concentrations use nominal solvent addition. **Naphthalene is available at HTEL; planned charge: 60 mg.**

## Reaction stocks

| Stock | Preparation target at 100% assay | Per reaction |
|---|---|---|
| Naphthalene source | **60 mg solid in 1 C2**; robot adds ~936 µL toluene for **500 mM**, adjusted to actual mass/assay. | Used to make stocks below |
| SM + naphthalene | **~322.32 mg SM /2 mL: 1 M SM +100 mM IS**. Robot adds **400 µL source +1600 µL toluene** at the nominal SM charge. | **25 µL** |
| Each ligand | **20 mM toluene**; weighed charges/reserves in CSV. Robot prepares bulk, mixes and fills working tubes. | **75 µL each**; reference 150 µL L17 |
| Pd(COD)(DQ) | Direct in **4 A3/B3/C3: 22.8 mg each; A4: 21.6 mg**. Robot adds ~1.50/~1.43 mL toluene for **40 mM**. Total **90 mg**, ~5.94 mL; **10 mg for scope**. | 75 µL |
| RockPhos L03 | **~800 µL /20 mM**, 7.50 mg pure equivalent; 7.73 mg at 97%. Prepared directly in **1 C1**; retain that tube. **10 mg available, max 4 R2 wells.** | 75 µL |

**Retain working tubes:** SM/IS **4 D2 starts at 1.89 mL** for both rounds; all four Pd tubes and RockPhos **1 C1** stay in use. No SM/Pd refill. Follow the experiment and source ledgers for routing/losses.

## Robot-made workup fluids

Each hold: robot makes 10% aqueous FA, then fluids in reservoir 2. Coarse-load raw solvents per CSV into A1-H1 reservoir in temporary slot 3.

| Fluid | Composition | Use |
|---|---|---|
| Quench | **9:1 ACN/water, 0.15% v/v FA**, no IS | 500 µL; open vials, heater off, 750 rpm /2 min |
| LC diluent | **1:1 water/ACN, 0.1% v/v FA**, no IS | 980 µL +20 µL extract |

## Robot-made authentic standards

Weigh separate FIT and CHECK charges per analyte: **at least 5 mg**, assay-corrected; enter actual masses. Weigh liquid product by difference. Robot makes **10 mM** masters (SM ~6.45 mg /4 mL; product ~5.83 mg /6 mL), matched matrix and **0.5/5 mM FIT dilutions**. Retain reserves.

Robot makes **8 mL /2 mM analytical IS** from **32 µL of 500 mM source**, crediting carrier toluene toward the matched matrix; loads **1 mL into analytical A3**. `IS_MASTER` starts empty; weigh only `IS_SOURCE`. Solid naphthalene: CAS 91-20-3; MW 128.17.

## Calibration during hold one - 14 injections

Robot fills/pipette-mixes **CALIBRATION**: SM A1-G1, product A2-G2. Seal/move directly to LC during R1. Follow generated preparation and injection maps.

| Analyte, µM | Role | Stock, mM | Analyte, µL | Matrix, µL |
|---|---|---|---|---|
| 0 | Fit blank | None | 0 | 966.667 |
| 10 | Fit | 0.5 | 20 | 946.667 |
| 50 | Fit | 0.5 | 100 | 866.667 |
| 150 | Fit | 5 | 30 | 936.667 |
| 350 | Fit | 5 | 70 | 896.667 |
| 800 | Fit | 5 | 160 | 806.667 |
| 666.667 | Independent check | 10, CHECK | 66.667 | 900.000 |

Add **33.333 µL IS per row**. Fit blank +five levels; exclude CHECK. Require **$R^2$ >=0.995**, **check error <=15%**, fit error **<=15%** (**<=20%** at 10 µM), measured separation/LOQs. Quantitative floor: max(LOQ, lowest nonzero calibrator). Python calculates yields.

# 03 / Local pipeline

Official **48 pairs (24 consensus +24 Sobol)**: `campaigns/hackathon2026/round1_design.csv`. Already prepared; no GPU or paid R1 rerun. Python quantifies UV data; cached Claude scoring selects R2. **Regenerate stocks/protocols/maps after updating the repository; earlier R1 exports use different pairs.** Physical actions: bench sheet.

## Install and configure

Python 3.12: `uv sync --locked --python 3.12 --extra sources --extra robot`; prefix commands with `uv run`. Or install `.[sources,robot]` in a venv. Copy config to `local_data/campaign.json`, calibration template to `local_data/calibration-plan.csv`, key example to `anthropic.key`. Put the raw Claude key there; preserve existing keys. Never send it to the robot/Git.

Use schema **6** and naphthalene IS; replace older configs/protocols. Preserve confirmed hardware/method facts, **re-establish IS retention time and calibration with naphthalene**; old mesitylene calibration is invalid. Both open shakes are **750 rpm /2 min**. Regenerate preparation/protocol files; leave workup vials open through sampling, recap for storage. Configure method/channel, full cycle and RTs/LOQs. Choose nondetection policy: default `reject` stops a missing-analyte result; `censor` reports bounds after RT/LOQ validation. **Default bulk-source reach fails:** supply actual suitable short-container/rack geometry before live release. Concentrations use nominal solvent addition. Twelve hours is advisory; record real times.

## Robot preparation, then R1

Generate a guarded preview; fill actual masses/assays in `local_data/charges.csv` from its template. Pd: **22.8 mg each in 4 A3/B3/C3; 21.6 mg in 4 A4**, 90 mg total. **IS_SOURCE: 60 mg naphthalene in 1 C2; IS_MASTER empty.** Follow carrier/raw-load CSVs. After local checks:

```
python -m hte.cli --config local_data/campaign.json prepare-stocks \
  --output output/prep-preview
python -m hte.cli --config local_data/campaign.json prepare-stocks \
  --charges local_data/charges.csv --output local_data/stock-prep --live
```

Run `local_data/stock-prep/prepare_stocks.py` locally: p300 single left /p1000 single right. Preserve its manifest at the default configured path. Follow swaps; afterward change right to p300 multi and tips 8/9 to 300 µL through the App installation/calibration workflow.

```
python -m hte.cli --config local_data/campaign.json export \
  --design campaigns/hackathon2026/round1_design.csv \
  --output output/r1-dose --live --stage dose_and_react
```

Start with **48 NEW EMPTY R1 vials**. Robot adds 25 µL SM/IS per vial, then catalysts, and heats the capped block. Retain **SM/IS tube 4 D2**, capped between rounds; its 1.89 mL initial fill covers both rounds. Robot makes standards/workup fluids during the hold. Send calibration plate to LC using generated map. Import/calibrate; set `lc_calibration_validated=true` after success. Export R1 workup with the SAME command/design, changing to `--output output/r1-workup --stage workup`. Keep prepared reservoir 2; use a fresh final LC plate.

## R2 and finish

Complete actual injection maps. After ALL 48 results:

```
python scripts/hte-lc-results.py next \
  --dosing output/r1-dose/dosing.csv \
  --exports local_data/r1-batch*.csv \
  --injections local_data/r1-map*.csv --output output/r1-decision
```

Calibration defaults to `local_data/calibration.csv`. QC precedes Claude: **387 untested pairs →22 +L17 reference**, L03 max four. **After interruption rerun the SAME command/output**; accepted scores are reused. Inspect uncertain API requests; a new folder may repeat charges. Run printed `refill_stocks.py` **FIRST** for ligand top-ups; retain SM/IS **4 D2** and all four Pd tubes without refilling. Fit **23 NEW EMPTY vials** in the same block and a NEW EMPTY reservoir 2; R2 Python adds **25 µL SM/IS per vial, then catalysts**. If unreleased, export `next_design.csv` with R1 `--history`, `--live` and a new export folder. After 23 final injections, run helper `final`; no Claude.

## Return records

Keep actual weights, preparation/source ledgers, LC exports/maps, calibration/QC, Claude records and matching doses/protocols. Copy robot JSON to `local_data/robot_logs/`, raw LC/MS/method files to `local_data/instrument_raw/`, times/deviations to `local_data/run-notes.txt`. Stop writers; `python scripts/hte-return-package.py` creates `transfer/hte-results.zip` with hashes, credentials excluded. Send privately; Git does not preserve local run data.